# PHASE 1 REPORT

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## 1. Introduction

This report presents the design and implementation of a single-cycle 16-bit processor using Logisim. It supports R-type, I-type, and J-type instructions. The objective is to design a correct datapath and control unit.

## 2. Datapath Design

The datapath executes instructions through five stages: Fetch, Decode, Execute, Memory, and Write Back.

Main components include:
- Program Counter (PC)
- Instruction Memory
- Register File (R0–R7)
- ALU (Arithmetic, Logic, Comparator, Shifter)
- Data Memory
- Immediate Extender
- Multiplexers
- Next PC Logic

## 3. Control Unit Design

The control unit consists of Main Control and ALU Control.
Main Control generates signals based on opcode.
ALU Control refines ALU operation using ALUOp and funct.

### Control Signals Table

| Instruction | RegDst | RegWrite | ALUSrc | MemRead | MemWrite | MemToReg | BEQ | BNE | J | JAL | JR | LUI | ALUOp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R-type | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | R-type |
| ADDI | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | ADD |
| ANDI | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | AND |
| ORI | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | OR |
| SLTI | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | SLT |
| LW | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | ADD |
| SW | X | 0 | 1 | 0 | 1 | X | 0 | 0 | 0 | 0 | 0 | 0 | ADD |
| BEQ | X | 0 | 0 | 0 | 0 | X | 1 | 0 | 0 | 0 | 0 | 0 | SUB |
| BNE | X | 0 | 0 | 0 | 0 | X | 0 | 1 | 0 | 0 | 0 | 0 | SUB |
| J | X | 0 | X | 0 | 0 | X | 0 | 0 | 1 | 0 | 0 | 0 | X |
| JAL | R7 | 1 | X | 0 | 0 | PC+1 | 0 | 0 | 1 | 1 | 0 | 0 | X |
| JR | X | 0 | X | 0 | 0 | X | 0 | 0 | 0 | 0 | 1 | 0 | X |
| LUI | 0 | 1 | X | 0 | 0 | LUI | 0 | 0 | 0 | 0 | 0 | 1 | X |

## 4. Testing and Simulation

To test the processor, a provided test case file from Blackboard was used. The corresponding text file containing the machine instructions was loaded into the instruction memory (ROM) in Logisim. The simulation was then executed to verify the behavior of different instruction types. The results showed that all instructions produced the expected output.

## 5. Conclusion

The processor successfully executes all instructions correctly. The design is ready for pipelining with minor refinements.